DOI: https://doi.org/10.5114/pm.2025.152241

Menopause Rev 2025; 24(2): 131-136

REVIEW PAPER

Position paper of the expert panel of the Polish Society of Menopause and Andropause on the use of Oestrogel® in menopausal hormone therapy*

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*The expert panel of the Polish Society of Menopause and Andropause, composed of Małgorzata Bińkowska (Warszawa), Tomasz Paszkowski (Lublin), Violetta Skrzypulec-Plinta (Katowice), nd Wojciech Zgliczyński (Warszawa), performed a detailed analysis of the literature on the use of Oestrogel® in menopausal hormone therapy.

Abstract

This position paper of the expert panel presents a comprehensive review of the efficacy, safety, and clinical application of bioidentical hormone replacement therapy (HRT), with particular focus on transdermal 17β-oestradiol gel (Oestrogel®). Bioidentical hormones – chemically identical to endogenous human hormones – are increasingly recognised as the preferred option in modern HRT, consistent with current international guidelines. Based on a review of randomised clinical trials and observational studies (sourced primarily from PubMed and Medline), transdermal oestradiol demonstrates a superior safety profile compared to oral formulations. Benefits include stable serum oestradiol levels without supraphysiological fluctuations; minimal impact on hepatic synthesis of procoagulant factors, triglycerides, hormone-binding proteins (SHBG, TBG, CBG), and inflammatory mediators (e.g. C-reactive protein); no increased risk of venous thromboembolism or ischaemic stroke; and the ability to use lower doses while maintaining efficacy. Additional advantages are a more physiological E2/E1 ratio, reduced inter-individual variability, and the option to monitor serum oestradiol levels. Oestrogel® supports dose flexibility and personalisation, allowing treatment to be tailored according to patient needs and guideline-based therapeutic schemes. When combined with micronised progesterone – the gold-standard progestogen for endometrial protection and the preferred option due to its favourable overall safety profile – this form of HRT offers a modern, well-tolerated, and individualised approach to the management of menopausal symptoms and osteoporosis prevention.

Key words: menopausal hormone therapy, bioidentical hormones, body-identical hormones, transdermal route of delivery.

Introduction

The search for an optimal menopausal hormone replacement therapy (HRT) that is both effective and safe and at the same time convenient to use has continued since the introduction of the first hormonal products to alleviate menopausal symptoms 80 years ago. Accumulating evidence suggests that bio-identical HRT, i.e. 17β oestradiol (E_2) and micronised progesterone (MP), has the most favourable benefit-risk ratio [1].

Bio-identical hormones are substances that have the same chemical and structural structure as endogenous hormones produced in the human body. In HRT, the term was popularised by Panay *et al*. [2], who termed bio-identical hormones as precise duplicates of oestradiol, progesterone, and testosterone, as physiologically synthesised in women's ovaries. The term "body-identical hormones", which is sometimes used interchangeably, can also be found in the medical literature. It seems, however, that in Polish language the term "bio-identical hormones" is simpler and understandable.

The classification of steroid hormones used in HRT was clearly presented by Stańczyk *et al*. [3], who divided them into 4 classes. Class A includes naturally occurring substances that have found their use as a medicine without modification of their chemical structure. This class includes conjugated equine oestrogens obtained from the urine of pregnant mares, introduced into treatment in the 1940s (in 1941 in Canada and in 1942 in the USA).

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Submitted: 07.02.2025
Accepted: 29.05.2025

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Class B includes bio-identical hormones derived from steroids, i.e. diosgenin and stigmasterol, obtained through a technological process from plants (for example Mexican potato, soya beans), and thus substances present in nature but requiring biosynthesis. From a physiological point of view, there is no apparent evidence to indicate that the bio-identical hormones obtained by biosynthesis show any difference from the clinical outcomes from the endogenous ovarian hormones. However, it should be noted that the female body does not have enzymatic systems that would enable it to obtain steroid hormones from these precursors.

Class C hormones are produced in a complex technological process using non-steroidal precursors. An additional difference from class B is also the fact that the process yields a mixture of even more than 10 isomers, only one of which is identical to the endogenous ovarian hormone.

Class D comproses completely synthetic hormones, which have no equivalent in nature, even though they can be produced from plants. It includes ethinyloestradiol and progestins such as dienogest, drospirenone, medroxyprogesterone acetate, or norethisterone acetate, present in HRT products.

Class B bio-identical hormones are available as medicines approved by the Food and Drug Administration and European Medicines Agency on the basis of the results of rigorous studies covering production, safety, and efficacy.

In light of publications from the last 2 decades, and well-established use for more than 40 years supported by the most recent updated international guidelines, including those from the European Menopause and Andropause Society, North American Menopause Society, International menopause Society, US Endocrine Society, British Menopause Society, and National Institute for Health and Care Excellence, transdermal oestrogen therapy has the best safety profile, with the following advantages [4–8]:

- bypassing the hepatic first-pass effect, which results in the following:
  - more constant and consistent bioavailability regarding absorption versus oral route, more stable serum oestradiol concentrations without its supraphysiological fluctuations (peaks and valleys) [9],
  - less effect on hepatic synthesis of procoagulant factors, triglycerides, hormone-binding proteins (SHBG, TBG, CBG), and inflammatory mediators (C-reactive protein);
- no increased risk of venous thromboembolic complications in at least 5 independent observational studies in France and the UK;
- no increased risk of ischaemic stroke based on observational studies;
- the possibility to individualise the treatment by using lower or higher doses of oestradiol compared to other routes of administration for oestrogen;
- a more physiological E_2/E_1 (oestradiol/oestrone) ratio, corresponding to the proportions of the premenopausal period;
- the possibility to monitor E_2 blood levels (only recommended in the case of suboptimal clinical response, especially in women using high doses) or when women with no or minimal menopausal symptoms want hormone therapy for bone protection and to confirm that levels are therapeutically efficacious [10].

The growing interest in well-tolerated transdermal oestradiol gel formulations is understandable in this context.

Product characteristics

Oestrogel® is a clear to slightly opalescent transdermal gel. One gram of gel contains 0.60 mg of oestradiol hemihydrate, which is chemically and biologically identical to the endogenous human hormone. The lowest available dose of Oestrogel® released by one pump actuation is 1.25 g of gel, equivalent to 0.75 mg E_2 [11].

According to the Summary of Product Characteristics, Oestrogel® is indicated for use in postmenopausal women as follows:

- HRT for oestrogen deficiency symptoms in postmenopausal women,
- prevention of osteoporosis in postmenopausal women at high risk of future fractures.

Pharmacodynamic properties

Transdermal administration of E_2 in the form of gel has different pharmacokinetic and pharmacodynamic parameters compared not only to oral administration of this hormone, but also to its application through a patch [9].

Pharmacokinetic studies indicate that the product needs to be applied topically (transdermally) to a large area of the skin in a volatile hydro-alcoholic solvent, and approximately 10% of oestradiol is transdermally absorbed into the bloodstream by passive diffusion through the different layers of the skin, which plays the role of a reservoir, unlike a patch, regardless of the patient's age. Daily application of 1.25–5 g of gel to a skin surface area of at least 750 cm^2 causes a gradual increase in blood oestrogen levels, which reach a steady state after about 3–5 days, and ensures circulating blood oestradiol and oestrone concentrations corresponding to those occurring in the early follicular phase of the natural menstrual cycle, in terms of both absolute values and the E_2/E_1 ratio [9].

The mean oestradiol and oestrone levels after 12 weeks of treatment with Oestrogel® at a dose of1.5 mg E_2/day were 65 and 58 pg/ml, respectively, compared to 33.5 and 49.0 pg/ml with 0.75 mg E_2/day [12] (Table 1).

Table 1. Oestradiol dose relationships depending on the route of administration and product type [13]

Product	Ultra-low	Low	Average	Higher	High
Oestrogel® (pump)	½	1	2	3	4^
Spray (dose)	1	2	3	4–5*	6*
Transdermal patch [µg]	12.5	25	50	75	100
Oral tablet [mg]	0.5	1	2	3^	4^

*Off-label use.
^Rarely required for symptom control in physiological menopause.

The half-life of oestradiol from Oestrogel® is 36 hours.

The optimal E_2 concentration ensuring the resolution of vasomotor symptoms and the prevention of osteoporosis is 60–150 pg/ml (220–550 pmol/l). In fact, an E_2 level of only 60 pg/ml (220 pmol/l) already suppresses vasomotor symptoms in 50% of women and prevents bone resorption. At an E_2 level reaching 100 pg/ml (400 pmol/l), hot flushes are reported to be abated in 100% of women and, of course, bone mineral density is protected [10].

Inter-individual variability in oestradiol pharmacokinetics results in up to a 10-fold difference in E_2 concentrations achieved with the same dose of oestradiol in the patch or gel. It is estimated that in about 20% of women transdermal absorption is worse than in others – they are the so-called "poor absorbers". This problem can be often overcome by increasing the daily dose of gel. The ability to control the achieved oestradiol concentrations is an additional advantage of the transdermal route of administration.

Literature review on the efficacy and tolerability profile of Oestrogel® in menopausal hormone replacement therapy

The optimal benefit/risk ratio of an E_2 transdermal gel (Oestrogel®) in the management of symptomatic menopausal women has been well-established, with more than 40 years of clinical experience since it was approved for the first time in 1972 in France. In a randomised placebo-controlled study, Jensen *et al.* [14] demonstrated the efficacy of Oestrogel® at a dose of 3 mg E_2/day in combination with oral micronized progesterone (MP; 200 mg for 12 days), as combined sequential hormone therapy in symptomatic postmenopausal patients. A highly significant reduction in the severity of menopausal symptoms was obtained, accompanied by an oestrone/oestradiol ratio close to unity, which is a value characteristic of the physiology of the premenopausal period.

Since then, many RCTs have been conducted to confirm the efficacy of Oestrogel® for the management of symptomatic menopausal women.

Dupont *et al.* [15] compared the clinical and endocrine effects of treatment with Oestrogel® at a dose of 1.5 mg E_2/day with or without micronised oral progesterone vs. a combined formulation of conjugated oestrogens and medroxyprogestone acetate in natural or surgical postmenopausal patients. Transdermal oestrogen therapy with gel has proven to be highly effective and to have a superior safety profile vs. the comparator product. Treatment with Oestrogel® had a neutral effect on liver function markers, serum angiotensinogen concentration, and the serum E2/E1 ratio. Similar results were obtained by Kornafel *et al.* [16], who analysed the efficacy and tolerability profile of monotherapy with Oestrogel® at a dose of 1.5 mg E_2/day.

Foidart *et al.* [17] conducted a randomised multicentre study in which patients received oestradiol gel (Oestrogel®) 1.5 mg daily and nomegestrol acetate (5 mg/day) or placebo as a combined sequential therapy. A significant decrease in the severity of menopausal symptoms according to the Kupperman index, favourable tolerability profile, and good cycle control were observed in the oestradiol gel group. Endometrial biopsy showed no evidence of hyperplasia in any of the HRT recipients.

Devogelaer *et al.* [18] conducted a long-term, randomised, double-blind study in hysterectomised patients receiving Oestrogel® at a daily dose of 1.5 mg E_2 versus a control group treated with estriol 2 mg/day oral tablet. These authors observed effective reversal of bone loss in patients receiving the transdermal E_2 gel compared to the control group.

The efficacy and tolerability of transdermal oestrogen therapy administered with a patch and gel in tropical climatic conditions were compared. Studies conducted in Brazil and Thailand showed greater efficacy in reducing menopausal symptoms and fewer local cutaneous adverse reactions in the Oestrogel group® compared to the patch group [19, 20].

In a pivotal randomised phase 3 placebo-controlled study to gain approval in the US, Archer [21] evaluated the efficacy and tolerability of transdermal HRT with Oestrogel® at doses of 0.75 mg E_2/day and 1.5 mg E_2/day, versus placebo. The participating women were either physiologically or surgically postmenopausal, and they were followed up for 12 weeks. The efficacy in the reduction of vasomotor menopausal symptoms was statistically significantly higher compared to placebo for both doses of the E_2 gel, although it was more pronounced in patients receiving 1.5 mg E_2/day. A similar tolerability profile was observed in both groups receiving the E_2 gel,

although oestrogen-related adverse reactions were observed more frequently in the higher-dose group. In the conclusion of that study, the author states that the investigated product is a well-tolerated and effective method of alleviating menopausal vasomotor symptoms and suggests that treatment should start with 0.75 mg E_2/day and the dose may be increased when required to 1.5 mg E_2/day.

Archer *et al.* [12] attempted to determine the minimum effective dose of oestradiol administered through transdermal gel in the treatment of symptomatic postmenopausal patients. In a first phase 3 study, patients were randomised to receive either Oestrogel® (1.5 mg or 0.75 mg E_2/day) or placebo. On the other hand, in a phase 4 study, randomised patients received daily doses of oestradiol 0.375 mg, 0.270 mg, or placebo. Efficacy in relieving vasomotor symptoms and improving cytological markers of vaginal atrophy was demonstrated in a 12-week follow-up at doses of 1.5 and 0.75 mg E_2/day. The results from the 2 studies supported the dose of 0.75 mg E_2 gel being the lowest practical dose for the treatment of symptomatic postmenopausal women. In conclusion, the authors of that paper state that the minimum effective dose of transdermal gel in Oestrogel® is 0.75 mg E_2/day, which enables a significant reduction of vasomotor symptoms and an improvement in the cytoarchitecture of the vaginal epithelium.

In a randomised placebo-controlled study, Yoon [22] demonstrated that in women with a history of osteoporotic hip fracture, the efficacy measured as the prevention of recurrent fractures and overall mortality at 4 years was similar in the group receiving risendronate (35 mg QW) and in the group receiving oestradiol transdermal gel (Oestrogel®; 1.5 mg E_2/d) combined with oral MP (100 mg/d) in continuous therapy. Bone mineral density of the femur did not increase over 4 years of follow-up in risendronate-treated women, whereas it increased significantly (by 2.8%) in the group receiving transdermal oestradiol gel and oral MP.

Mizunuma [23] evaluated the efficacy and safety profile of low-dose maintenance oestrogen therapy using transdermal oestradiol gel in women experiencing vasomotor symptoms of oestrogen deficiency. Those patients received transdermal oestrogen therapy in the form of gel at a dose of 1.08 mg E_2/d with or without a progestogen (continuous or sequential medroxyprogesterone acetate) for 8 weeks. After a satisfactory effect in the form of relief of vasomotor symptoms was obtained, the patients were randomised to receive the gel at a dose of 0.54 mg E_2/d or placebo for another 16 weeks. In the group receiving the maintenance dose of oestradiol, clinically satisfactory maintenance of the therapeutic effect was observed, with good tolerability and a high level of patient satisfaction.

Piette [24] presented the results of studies evaluating the efficacy and tolerability of transdermal therapy with Oestrogel® at a daily dose of 0.75 or 1.5 mg E_2 compared to a transdermal patch releasing 50 mcg E_2/day, both in combination with oral MP, at the 19th World Congress on Menopause in Melbourne (October 2024). In a 12-week follow-up, similar efficacy in relieving vasomotor symptoms was observed for both doses of the gel and for the transdermal patch. All treatments induced a dose-dependent increase in oestradiol levels and resulted in an oestrone/oestradiol ratio close to unity. A significant difference was found in the frequency of cutaneous adverse reactions occurring at the drug application site. In the case of the patch applied once a week, these symptoms occurred in 23% of women, compared to only 1.1% of women using the gel. The authors of that study recommend personalised HRT with regard to the oestradiol dose and route of administration.

If progestogen needs to be used with Oestrogel® to protect the endometrium, it is important to consider the significant differences in safety profile among the different progestogens. Oral MP appears to be the most favourable in this context in clinical trials [7, 8] (PEPI 1995). The following MP doses are recommended during combined HRT with standard oestrogen doses [13, 25]:

- 200 mg/day orally for 12 days (sequential combination therapy),
- 100 mg/day orally (continuous combination therapy) (Table 2).

Several papers have been published suggesting the benefits of using the levonorgestrel-releasing intrauterine system (LNG-IUS 20) to protect the endometrium during combined HRT [5, 26]. These benefits include not only efficient endometrial protection against the proliferative effects of oestrogen but also a favourable safety and tolerability profile of intrauterine progestogen therapy. The use of LNG-IUS20 during transdermal oestrogen gel therapy seems particularly beneficial [26, 27]. Combined continuous HRT with Oestrogel® (1.5 mg E_2/d) and LNG-IUS (52 mg) appears to be effective and well tolerated*. However, the condition for endometrial safety is the correct positioning of LNG-IUS 20 in the uterine cavity (assessment by transvaginal ultrasonography is recommended: not lower than 2 cm from the uterine fundus, without the IUS sliding into the cervical canal) [13].

The literature on the subject includes publications indicating the benefits of transdermal oestrogen therapy in perimenopausal patients who suffer from migraines. Transdermal oestradiol appears to be a preferable option in these patients compared with oral oestrogen because of more stable serum oestradiol levels [28, 29]. Owing to the relatively low risk of ischaemic stroke with transdermal oestradiol and oral MP therapy, it is recommended that this type of HRT be preferentially used in patients experiencing migraine

*An off-label use in Poland.

Table 2. Recommended oral doses of micronised progesterone depending on the dose of oestradiol and on the type of therapy

Oestradiol dose	Micronised progesterone	
	Sequential therapy	Continuous therapy
Ultra-low/low [mg]	200	100
Average [mg]	200	100
Higher [mg]	200	100
High [mg]	200–300	100–200

attacks with aura [30]. Moreover, transdermal oestrogen gel therapy has been shown to be effective in preventing menstrual migraine attacks. MacGregor *et al.* [29] obtained promising effects in prevention of migraine attacks using a dose of 1.5 mg E_2/day in gel (Oestrogel®) from cycle day 10 to the second day of menstrual bleeding, although the therapeutic effect was reversed after discontinuation of oestrogen therapy. More studies are needed, to avoid or attenuate this rebound effect, i.e. by prolonging the treatment for up to 7 days after cessation of menstrual bleeding.

A review of the literature on the use of Oestrogel® in menopause suggests that it is a well-tolerated formulation with high efficacy in the relief of menopausal symptoms, which ensures a favourable safety profile of HRT when used alone or in combination especially with MP. The potential for dynamic adjustment of both the oestradiol dose and the treatment regimen to achieve an individual therapeutic objective is compatible with the modern personalisation trends in menopause medicine.

Practical notes:

1. Each dose (one pump) of Oestrogel® contains 0.75 mg E_2.
2. The first dose from each package should be discarded.
3. Each pack contains 60 doses.
4. The application surface area is at least 750 cm^2, which corresponds to the entire arms from wrist to shoulder and/or the inner part of the thighs).
5. The gel should not be applied to the breast or vulva due to the alkohol content.
6. The application site should be left to dry for 5 minutes before covering with clothes.
7. Hands should be washed after applying the gel.
8. Creams, lotions, or other cosmetics should not be applied to the area where the gel has been applied for 1 hour.
9. The dose ensuring effective relief of vasomotor symptoms in most women is two metered doses(2 pumps), i.e. 1.5 mg E_2.
10. If satisfactory control of menopausal symptoms is not achieved, it is recommended that the method and place of application (using a product model containing placebo) be controlled and the E_2 concentration achieved in the blood be tested.
11. The dose may be increased gradually up to a maximum of 4 pumps, i.e. 3 mg E_2, remembering about the need to increase the dose of MP or another progestogen used in women with an intact uterus.

Conclusions

Oestrogel® is a transdermal gel containing bioidentical oestradiol.

It is used in HRT to treat menopausal symptoms caused by oestrogen deficiency. The second indication is prevention of osteoporosis in postmenopausal women at high risk of bone fractures.

It is used as monotherapy in hysterectomised women.

In women with an intact uterus, it is necessary to add a progestogen at an adequate dose.

One dose, i.e. one gel pump applied to the skin, delivers 0.75 mg E_2.

The therapeutic range is 1–4 doses, i.e. 0.75–3.0 mg E_2.

The standard dose is 1.5 mg E_2, i.e. 2 gel pumps.

Disclosures

1. Institutional review board statement: Not applicable.
2. Assistance with the article: None.
3. Financial support and sponsorship: None.
4. Conflicts of interest: None.

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